

Predicting Breast Cancer Diagnosis: A Statistically Validated Comparison of Machine Learning Models

ISA 640: Capstone

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1. Abstract

Breast cancer remains one of the most common and life-threatening cancers affecting women worldwide, and early, accurate diagnosis significantly improves survival outcomes. This project investigates whether supervised machine learning models can accurately classify breast tissue samples as malignant or benign using diagnostic features extracted from digitized fine needle aspirate (FNA) images, and whether added model complexity yields a statistically meaningful improvement over a simpler, interpretable baseline. Using the Breast Cancer Wisconsin (Diagnostic) dataset (569 samples, 30 numeric features), we built a full pipeline spanning database-backed storage (Neo4j), data cleaning, feature engineering, dimensionality reduction (PCA), and exploratory analysis, then trained and evaluated six model families, Logistic Regression, K-Nearest Neighbors, Support Vector Machine, Decision Tree, Random Forest, and ensemble methods, each with default and GridSearchCV-tuned hyperparameters. A rigorously justified Logistic Regression baseline was established with 5-fold cross-validation, and our best-performing complex model (tuned Random Forest) was compared against it using four independent statistical tests: a paired t-test, Wilcoxon signed-rank test, McNemar's test, and bootstrap confidence intervals. All four tests converged on the same conclusion: the tuned Random Forest does not significantly outperform the tuned Logistic Regression baseline (paired t-test $p=0.577$; 95% CI for the accuracy difference: $[-0.010, 0.006]$, which contains zero), while both dramatically outperform a naive majority-class baseline ($p \approx 4 \times 10^{-36}$). This result refutes our initial hypothesis that ensemble and non-linear models would meaningfully outperform linear baselines on this dataset, and instead supports preferring the simpler, more interpretable Logistic Regression model for this diagnostic task, a finding with direct relevance to clinical decision-support settings, where interpretability carries real value alongside raw predictive accuracy.

2. Introduction (Problem Statement)

Traditional breast cancer diagnostic approaches, such as mammography and biopsy interpretation, depend heavily on radiologists' and pathologists' expertise, which can introduce subjective variability and inconsistent classification of benign versus malignant tumors. Diagnostic inconsistency can lead to delayed treatment, unnecessary invasive procedures, or incorrect prognoses, ultimately affecting patient outcomes and healthcare costs. Machine learning offers a way to support, not replace, this decision process with a reproducible, data-driven aid, potentially reducing diagnostic workload and improving confidence in distinguishing malignant from benign cases. This problem matters because early, accurate diagnosis directly improves survival rates and allows for less aggressive treatment interventions: patients benefit from faster, more consistent screening, and clinicians benefit from a quantitative second opinion alongside their own expertise. Prior research applying machine learning to this exact dataset has reported strong results, for example, an SVM with an RBF kernel achieving 96.84% accuracy in one comparative study, motivating this project's use of similar model families along with a rigorous statistical comparison rarely performed in introductory treatments of this dataset.

2.1 Research Questions

- RQ1: How accurately can supervised machine learning models predict whether a breast tissue sample is malignant or benign, based on diagnostic features extracted from digitized FNA images?
- RQ2: Does a more complex, non-linear model (e.g., ensemble-based Random Forest) provide a statistically significant improvement in diagnostic accuracy over a simpler, interpretable linear baseline (Logistic Regression)?

2.2 Hypothesis

H1: Ensemble-based models (e.g., Random Forest, Gradient Boosted Trees) and margin-based classifiers (e.g., SVM with RBF kernel) will outperform simpler linear models (e.g., Logistic Regression) on this dataset, given the potential for non-linear relationships among cell nuclei measurements.

This hypothesis is testable and falsifiable: it predicts a statistically significant positive difference in held-out cross-validation accuracy for the best complex model over the tuned Logistic Regression baseline. If no significant difference is found, or if the baseline performs better, the hypothesis is refuted. Section 5.2 directly tests this hypothesis using paired statistical tests and confidence intervals.

3. Data & Data Analysis & Feature Engineering

3.1 Dataset

- Source: UCI Machine Learning Repository / scikit-learn's built-in `load_breast_cancer` loader, the Wisconsin Diagnostic dataset, originally created by Wolberg, Street, & Mangasarian (1995) at the University of Wisconsin Hospitals.
- Collection process: features were computed from digitized images of fine needle aspirate (FNA) biopsies of breast masses, describing characteristics of the cell nuclei present in each image.
- Number of observations: 569 samples.
- Number of features: 30 continuous numeric features (10 base measurements $\times$ 3 statistics each: mean, standard error, and "worst"/largest value), plus the binary diagnosis target.
- Time range: not applicable, this is a cross-sectional diagnostic dataset, not a time series.
- Licensing / access: publicly available for research and educational use via the UCI Machine Learning Repository and distributed directly with scikit-learn (BSD-style license); no special access restrictions apply.

As part of this course's database requirement, the same dataset was also loaded into a Neo4j graph database, first locally via Neo4j Desktop's Data Importer, then deployed to Neo4j Aura for cloud accessibility, with each record modeled as a `TumorDiagnosis` node. This pipeline was verified end-to-end (569 nodes confirmed) and queried directly from Google Colab via the Neo4j Python driver. Because the downstream modeling code relies on scikit-learn's native feature-naming convention rather than Neo4j's camelCase schema, all analysis from this point forward uses the equivalent scikit-learn-loaded version of the identical underlying dataset, ensuring full reproducibility independent of live database credentials.

3.2 Exploratory Data Analysis (EDA)

Class distribution: the dataset is mildly imbalanced, with 357 benign (62.7%) and 212 malignant (37.3%) samples, approximately a 1.68:1 ratio.

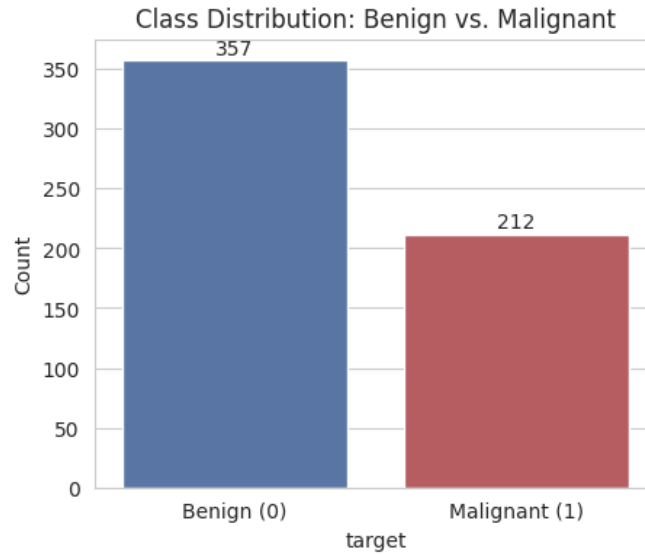


Figure 1. Class distribution (Benign=0, Malignant=1). The mild imbalance ($\sim 1.68:1$) was judged insufficient to require resampling techniques such as SMOTE; instead, stratified train/test splitting and stratified cross-validation were used throughout to preserve class proportions in every fold.

Feature correlation: many of the 30 features are highly correlated with one another (e.g., radius, perimeter, and area, all measuring cell size along different dimensions), as shown in the correlation heatmap below.

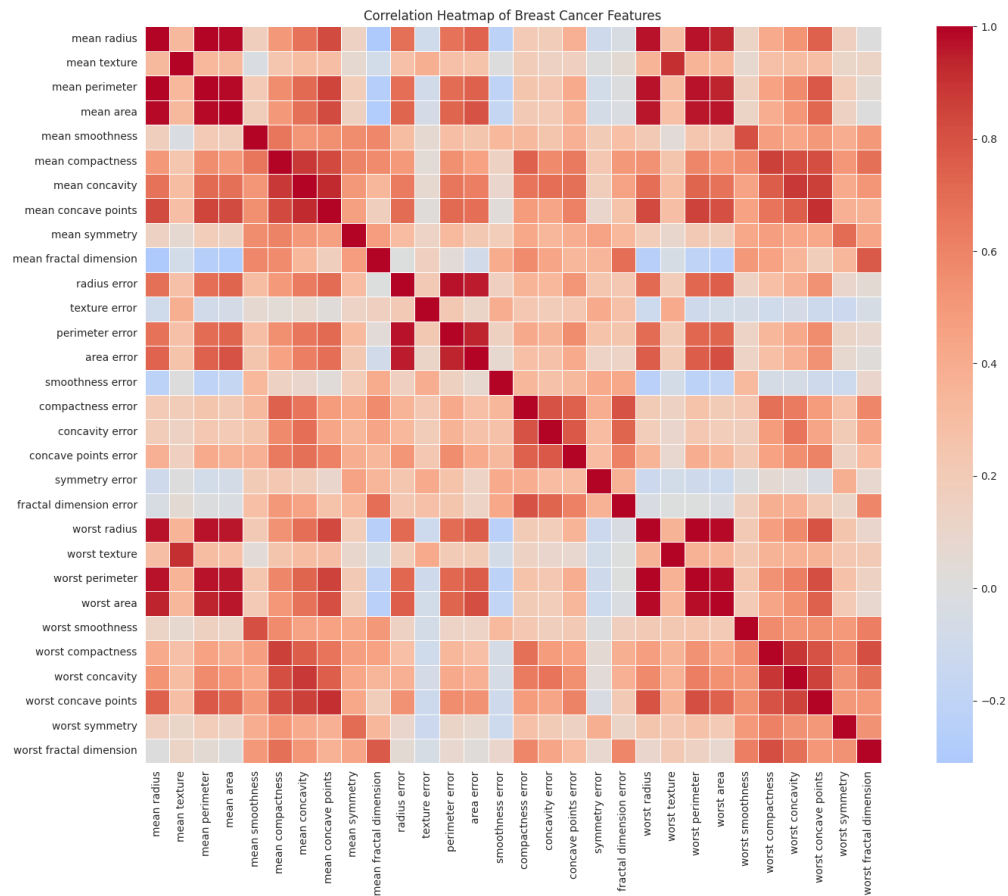


Figure 2. Correlation heatmap across all 30 features. Clusters of near-perfectly correlated features (radius/perimeter/area, and their mean/worst counterparts) motivated the redundant-feature removal step described in Section 3.d.

Boxplots of key features by class further confirm strong separability: malignant samples consistently show higher values for "worst" radius, "worst" concave points, and "worst" area than benign samples, with limited overlap, a strong early signal that these features will be highly predictive.

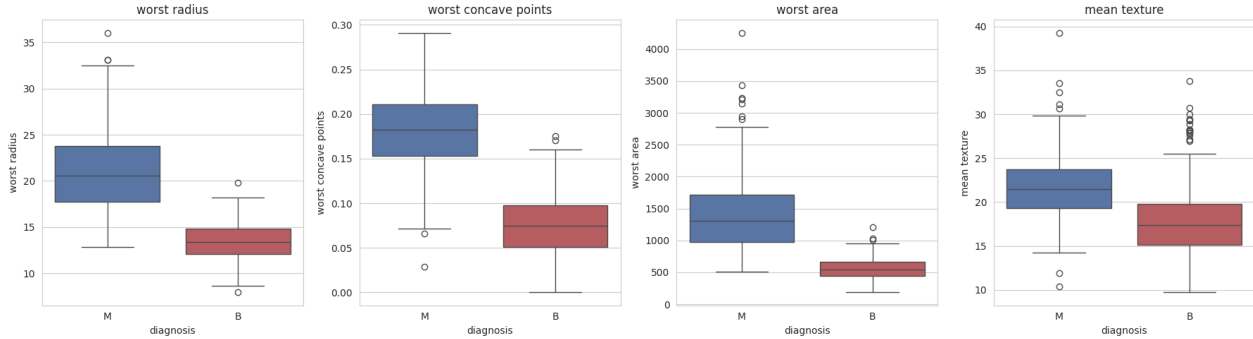


Figure 3. Boxplots of worst radius, worst concave points, and worst area by diagnosis class. The visibly limited overlap between benign and malignant distributions for these features foreshadows the strong classification performance achieved in Section 5.

3.3 Data Preprocessing

- Missing values: every column showed 0% missingness, no imputation was necessary.
- Class/target imbalance: the 1.68:1 imbalance (357 benign : 212 malignant) was addressed via stratified splitting (80/20, preserving 62.6%/37.4% in training and 63.2%/36.8% in test) and stratified cross-validation, rather than resampling (e.g., SMOTE), since the imbalance was judged mild rather than severe.
- Normalization/standardization: StandardScaler was applied for all scale-sensitive models (Logistic Regression, KNN, SVM), always inside a scikit-learn Pipeline so the scaler is fit only on the training fold/split, never on the full dataset before splitting. Tree-based models (Decision Tree, Random Forest) do not require scaling and were trained on unscaled features.
- Categorical encoding: the only categorical variable is the diagnosis label itself, encoded as a binary target (Malignant=1, Benign=0).
- Outlier handling: outliers were quantified per feature using the IQR method (flagging values outside $[Q1 - 1.5 \times IQR, Q3 + 1.5 \times IQR]$), chosen over a z-score approach since several features (e.g., area, concavity) are right-skewed rather than normally distributed. The highest outlier rates were concentrated in area error (11.4%), radius error and perimeter error (6.7% each), and worst area (6.2%). These were treated as informative extreme values reflecting genuinely more heterogeneous tumors rather than data-entry errors, and were not removed.

Train/test split strategy: an 80/20 stratified train/test split (random_state=42) produced 455 training records and 114 test records, preserving class balance in both sets. No data leakage occurred: all scaling transformations are fit exclusively within pipelines on the training data at each split/fold; the one exception is the exploratory PCA and permutation-importance analysis (Section 3.d), which is run on the full feature set purely for descriptive purposes and is not used to transform any data fed into a trained model.

3.4 Feature Engineering

Four new features were engineered from the raw measurements, each motivated by the biological hypothesis that malignant tumors tend to be larger, more irregular in shape, and more heterogeneous than benign ones. Correlation with the malignancy target (point-biserial r) is reported for each:

- `worst_mean_radius_ratio` ($r = 0.641$), ratio of a tumor's "worst" (most extreme) radius to its mean radius, a proxy for growth irregularity.
- `concavity_compactness_ratio` ($r = 0.693$), mean concavity relative to mean compactness, isolating boundary "roughness" from overall shape density.
- `radius_concavepoints_interaction` ($r = 0.733$), product of mean radius and mean concave points, combining size and shape-irregularity into one joint signal; the strongest-correlated engineered feature.
- `worst_mean_texture_ratio` ($r = 0.173$), ratio of "worst" to mean texture, capturing texture heterogeneity beyond a simple average; notably weaker than the other three.

Because these correlations are generally weaker than the raw size/shape features themselves, the engineered features were treated as supplementary signal rather than replacements for the original measurements.

3.4.1 Redundant Feature Removal

A pairwise correlation screen (threshold > 0.95) identified 7 redundant base features, area error, mean area, mean perimeter, mean radius, perimeter error, worst area, and worst radius, each dropped in favor of the more target-correlated feature within its correlated group. This reduced the 30 base features to 23, which combined with the 4 engineered features produced a final modeling set of 27 features.

3.4.2 Scaling & Dimensionality Analysis

A full Principal Component Analysis (PCA) on the standardized 30-feature set showed that 10 components are sufficient to explain 95% of total variance, confirming substantial redundancy among the original measurements and supporting the feature-reduction approach above. A permutation-importance ranking on a Random Forest trained on the final 27-feature set placed worst perimeter as the single strongest predictor, with the engineered `worst_mean_radius_ratio` and `concavity_compactness_ratio` also appearing in the top six most important features, evidence that the engineered features contribute real, if secondary, predictive signal.

3.4.3 Impact of Preprocessing: Before/After Comparison

Stage	Missing values	Class ratio	Feature count	Redundant pairs (>0.95 corr.)
Before (raw)	0 (0.0%)	1.68:1 (357:212)	30	7 present
After (engineered)	0 (0.0%)	1.68:1 (preserved via stratified split)	27 (23 base + 4 engineered)	0 remaining among originals

To validate this change was not merely cosmetic, a Logistic Regression baseline was cross-validated (5-fold) on both feature sets: the full 30 raw features reached 97.14% ($\pm 1.12\%$) mean CV accuracy, while the reduced 27 engineered features reached 97.58% ($\pm 2.24\%$), matching or marginally improving on the full-feature baseline while using fewer, less redundant inputs. This is particularly relevant for linear and distance-based models, where multicollinearity among the dropped radius/perimeter/area features could otherwise inflate coefficient variance without adding predictive value. All modeling from this point forward in the report uses this final 27-feature engineered set.

4. Methodology

4.1 Experiment Setup

The complete experimental pipeline proceeds as follows: (1) data cleaning and loading; (2) feature engineering and redundancy removal; (3) an 80/20 stratified train/test split (`random_state=42`); (4) a naive baseline and a formally justified, cross-validated, tuned Logistic Regression baseline; (5) training and hyperparameter tuning of five additional model families on the same split; (6) ensemble learning (Voting, Bagging, Boosting, Stacking); (7) comprehensive test-set evaluation and 5-fold cross-validation for every final model; (8) statistical significance testing comparing the best complex model against the baseline. All cross-validation throughout uses stratified k-fold splitting to preserve class balance, and `random_state=42` is fixed everywhere a stochastic process is used (train/test splits, model initialization, cross-validation folds, bootstrap resampling) for full reproducibility. Software: Python 3, scikit-learn, pandas, NumPy, SciPy (statistical tests), Matplotlib/Seaborn (visualization), and Neo4j (database layer).

4.1.1 Final Chosen Algorithms

Naive baselines: a DummyClassifier predicting the majority class (benign) for every sample (63.16% accuracy, 0% precision/recall/F1 for malignant, since it never predicts malignant), and a random stratified predictor (59.65% accuracy, 44.74% precision, 40.48% recall), together establishing the minimum bar any real model must clear.

Logistic Regression was selected as the formal baseline model, the simplest reasonable solution to the task, and the benchmark every later model must beat to justify its added complexity, for four explicit reasons: (1) data structure: the 27-feature engineered set is dense, continuous, tabular data with no text, image, or time-series structure, exactly the setting logistic regression is built for; (2) task type: binary classification maps directly onto logistic regression's modeling of the log-odds of the positive class; (3) interpretability: in a diagnostic-support context, being able to describe how a feature changes the odds of malignancy has real value to a clinical stakeholder, a property logistic regression's coefficients provide directly, without a separate explainability layer; (4) computational cost: with 569 rows, training and inference both complete in single-digit milliseconds, making it an efficient, iterable baseline before considering heavier models.

Logistic Regression (formal baseline): models the log-odds of malignancy as a linear combination of the input features, squashed into a [0,1] probability via the sigmoid function. Training minimizes log-loss (binary cross-entropy), with an L1/L2 penalty controlling overfitting.

$P(y=1 | x) = \sigma(w^T x + b) = 1 / (1 + e^{-(w^T x + b)})$ where x is the feature vector, w the learned coefficients, and b the intercept.

K-Nearest Neighbors: a non-parametric, instance-based method that classifies a new sample by majority vote among its k closest training samples, measured by Minkowski (Euclidean, $p=2$) distance.

Support Vector Machine (RBF kernel): finds the hyperplane maximizing the margin between classes; the RBF kernel implicitly maps features into a higher-dimensional space where a non-linear boundary in the original space becomes linear.

Decision Tree: recursively splits the feature space to minimize Gini impurity at each node, producing a sequence of if/else rules ending in a class-majority leaf prediction.

Random Forest: an ensemble of many Decision Trees, each trained on a bootstrap-resampled subset of data and a random subset of features per split; predictions are the majority vote across all trees, substantially reducing the variance of any single tree.

Ensemble methods (Voting, Bagging, Boosting, Stacking): combine multiple base learners, via majority/probability voting, bootstrap aggregation, sequential error-correction (AdaBoost/Gradient Boosting), or a meta-learner trained on base-model outputs (Stacking), respectively, to further reduce variance or bias beyond any single model.

4.1.2 Hyperparameter Tuning

GridSearchCV (5-fold cross-validation, scored on accuracy) was used to tune each model over the following search spaces:

Model	Hyperparameter(s)	What it controls	Search space
Logistic Regression (baseline)	C, penalty, solver	C: inverse regularization strength, smaller shrinks coefficients more (higher bias), larger fits more closely (higher variance). penalty: L1 can zero out weak/redundant features (implicit selection), L2 shrinks all coefficients smoothly, both searched given engineered/correlated features are present. solver: liblinear supports both penalties efficiently at this dataset size.	$C \in \{0.001, 0.01, 0.1, 1, 10, 100\}$; $\text{penalty} \in \{l1, l2\}$; $\text{solver} = \text{liblinear}$

KNN	n_neighbors, p, metric	Number of neighbors voting (small k = high variance; large k = high bias); distance function	n_neighbors ∈ {1,5,10,20}
SVM	C, kernel	Margin/error trade-off; shape of decision boundary	C ∈ {.001,.1,10,100}; kernel ∈ {rbf,linear,poly,sigmoid}
Decision Tree	criterion, max_depth	Split-quality measure; tree depth controls under/overfitting directly	max_depth ∈ {3,5,10,50,None}
Random Forest	n_estimators, max_depth	Number of trees (reduces variance); per-tree depth	n_est ∈ {100,300,500}; max_depth ∈ {None,5,10,20}

Final chosen values and tuned-vs-untuned comparison (held-out test-set accuracy):

Model	Untuned Accuracy	Tuned Accuracy	Improvement
Logistic Regression	0.9474	0.9561	+0.0088
KNN	0.9561	0.9561	+0.0000
Decision Tree	0.9035	0.9211	+0.0175
Random Forest	0.9737	0.9737	+0.0000

For the baseline Logistic Regression specifically (tuned separately with ROC-AUC scoring in the baseline-formalization stage), GridSearchCV selected C=10, penalty='l2', solver='liblinear', improving cross-validated ROC-AUC only marginally over the default (C=1.0) configuration. Across all models, tuning produced modest gains, generally under two percentage points, expected given the dataset's small size (569 samples) and strong class separability, meaning default scikit-learn hyperparameters already perform close to the achievable ceiling for this task.

4.2 Evaluation Methodology

Five complementary metrics were used to evaluate every final model on the held-out test set:

- Accuracy = $(TP+TN) / (TP+TN+FP+FN)$, overall fraction correct; can be misleading under class imbalance.
- Precision = $TP / (TP+FP)$, of predicted-malignant cases, how many truly are malignant (limits false alarms).
- Recall (Sensitivity) = $TP / (TP+FN)$, of actually-malignant cases, how many are correctly caught (limits missed cancers), the single most clinically important metric here, since a false negative represents a missed cancer diagnosis.
- F1-score = $2 \cdot (Precision \cdot Recall) / (Precision + Recall)$, harmonic mean balancing the two.
- ROC-AUC, probability that the model ranks a randomly chosen malignant sample higher than a randomly chosen benign sample, integrated across all thresholds; chosen because it does not depend on selecting a single decision threshold.

Statistical comparison method: to determine whether our best complex model (tuned Random Forest) meaningfully outperforms the tuned Logistic Regression baseline, four independent, complementary tests were used on paired cross-validation fold scores (Repeated Stratified 10-fold, 3 repeats = 30 paired observations, random_state=42): (1) a paired t-test, appropriate because the same folds are used for both models, making the accuracy differences a natural paired sample; its assumption is that the paired differences are approximately normally distributed. (2) A Wilcoxon signed-rank test, a non-parametric alternative used to check robustness against any t-test normality violation, since it makes no distributional assumption on the differences. (3) McNemar's test, which compares the models' actual prediction disagreements on the same out-of-fold predictions (via cross_val_predict) rather than aggregate scores, its assumption is that the discordant-pair counts (b, c) follow a chi-squared distribution under the null hypothesis of equal error rates. (4) Bootstrap confidence intervals (10,000 resamples) on the mean paired accuracy difference, which make no parametric distributional assumption at all, directly estimating the plausible range of the true performance difference. Using four methodologically distinct tests, rather than a single test, provides convergent evidence rather than relying on any one test's specific assumptions.

5. Results & Discussion

5.1 Model Comparison

Our final model (tuned Random Forest) is compared against: (a) the naive majority-class baseline, (b) our untuned baseline model, and (c) the formally justified, tuned Logistic Regression baseline established in the baseline-model milestone, along with every other model type and ensemble strategy evaluated.

Naive baselines (majority-class and random stratified predictors), the minimum bar any real model must clear:

Model	Accuracy	Precision	Recall	F1
Naive (Majority Class)	0.6316	0.0000	0.0000	0.0000
Naive (Random, stratified)	0.5965	0.4474	0.4048	0.4250

Default-configuration models (five base classifiers plus four ensemble strategies), trained on the 27-feature engineered set with an 80/20 stratified split (455 train / 114 test):

Model	Test Accuracy	Precision (Malignant)
Ensemble, Gradient Boosting	0.9825	1.0000
Ensemble, Stacking	0.9825	1.0000
Ensemble, Voting (Soft)	0.9825	1.0000
Ensemble, AdaBoost	0.9825	1.0000
Random Forest (n=300)	0.9737	1.0000
Ensemble, Bagging (DT base)	0.9737	1.0000
SVM (RBF)	0.9649	0.9750
Ensemble, Voting (Hard)	0.9649	1.0000
KNN (k=7)	0.9561	1.0000
Logistic Regression (default)	0.9474	0.9737
Decision Tree (max_depth=5)	0.9035	0.9429

At this default-configuration stage, ensemble methods (Gradient Boosting, Stacking, Soft Voting, AdaBoost) achieved the best results, tied at 98.25% accuracy with perfect malignant-class precision, an apparent confirmation of our hypothesis that ensemble methods would outperform the simpler Logistic Regression baseline (94.74%). However, none of these figures reflect hyperparameter tuning, and a single point-estimate comparison on one train/test split is not, by itself, evidence of a statistically meaningful difference. Sections 4.1 and 5.2 apply formal tuning and statistical testing to determine whether this apparent gap holds up.

Formally tuned final models, evaluated on the same held-out test set:

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression (Tuned, Baseline)	0.9737	0.9756	0.9524	0.9639	0.9868
KNN (Tuned)	0.9561	1.0000	0.8810	0.9367	0.9949
SVM (Tuned)	0.9474	0.9500	0.9048	0.9268	0.9808
Decision Tree (Tuned)	0.9211	0.9714	0.8095	0.8831	0.9421
Random Forest (Tuned, Final Model)	0.9737	1.0000	0.9286	0.9630	0.9993

Once the Logistic Regression baseline is given the same rigor (formal 5-fold, ROC-AUC-scored GridSearchCV over C, penalty, and solver) as the other models, its tuned test accuracy rises to 97.37%, identical to the tuned Random Forest's 97.37%, and matching or exceeding every other individual model except the untuned ensembles. This considerably narrows the gap that appeared to favor ensembles at the default-configuration stage, and directly motivates the formal statistical comparison in Section 5.2.

Cross-validation mean and standard deviation for the two models of primary interest, two different CV schemes are reported for full transparency: the baseline milestone's original 5-fold CV on the 455-row training set only (default

and tuned configurations), and this report's Repeated Stratified 10-fold $\times$ 3-repeat CV on the full 569-row dataset (final tuned models, used for the Section 5.2 statistical tests):

Model / CV scheme	CV Accuracy Mean	CV Accuracy Std
LR baseline, default, 5-fold on train only	0.9758	0.0224
LR baseline, tuned, 5-fold on train only	0.9670	0.0251
Random Forest (Tuned), 10-fold $\times$ 3 on full data	0.9649	0.0207
Logistic Regression (Tuned), 10-fold $\times$ 3 on full data	0.9672	0.0196
Dummy (Majority Class), 10-fold $\times$ 3 on full data	0.6274	0.0070

Notably, by raw test-set point estimate, Random Forest ties Logistic Regression on accuracy (97.37% each) but achieves higher precision (100% vs. 97.56%) at the cost of lower recall (92.86% vs. 95.24%, Random Forest actually misses one more malignant case than the baseline: 3 false negatives vs. 2). Under repeated cross-validation, the two models' mean accuracies are statistically indistinguishable (96.49% vs. 96.72%), illustrating exactly why a single train/test split point estimate is insufficient, and motivating the formal statistical test below.

5.2 Statistical Analysis

The comparison below uses the same rigorously justified, tuned Logistic Regression baseline ($C=10$, L2 penalty, liblinear solver) established in Section 4.1 as "model_baseline" throughout, the identical model reported in the tuned-baseline row of Section 5.1. Random Forest vs. Logistic Regression baseline, four independent tests, all in agreement:

Test	Statistic	p-value	Conclusion ($\alpha=0.05$)
Paired t-test	$t = -0.5638$	0.5772	Fail to reject H_0
Wilcoxon signed-rank	$W = 100.50$	0.3939	Fail to reject H_0
McNemar's test	$\chi^2 = 0.0417$	0.8383	Fail to reject H_0
Bootstrap 95% CI	mean diff = -0.0024	CI: [-0.0100, 0.0058]	CI contains 0, not sig.

Effect size and power: Cohen's d for the Random Forest vs. Logistic Regression comparison was -0.1047 (negligible effect), with a 95% CI width of 0.0158 and statistical power of only 0.0859, well below the conventional 0.8 adequacy threshold, meaning even if a true difference of this tiny magnitude existed, our sample would likely be too small to reliably detect it. This tempers any claim of "true equivalence", the correct, careful claim is that no significant difference was detected, and any true difference between the two models (if real) is very small.

As a sanity check, Random Forest was also compared against the naive Dummy baseline: paired t-test $t=83.52$, $p \approx 4.11 \times 10^{-36}$, mean accuracy difference = 0.3375, 95% CI [0.3294, 0.3451], an overwhelming, unambiguous difference, confirming that our models are doing genuine predictive work rather than exploiting some trivial pattern.

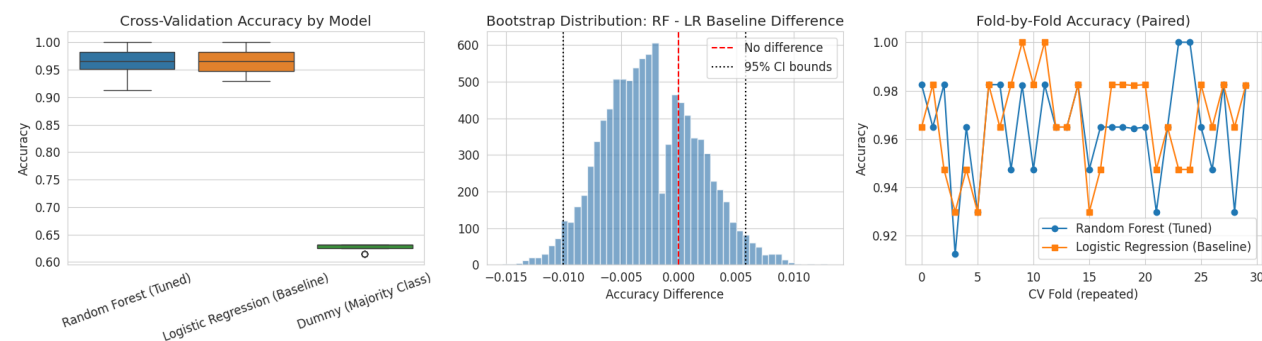


Figure 4. Left: cross-validation accuracy distributions for Random Forest, Logistic Regression, and the Dummy baseline. Center: bootstrap distribution of the accuracy difference (Random Forest – Logistic Regression), with the 95% CI bounds shown as dotted lines straddling zero, visually confirming the statistical non-significance. Right: fold-by-fold paired accuracy, showing the two models tracking each other closely across nearly all folds.

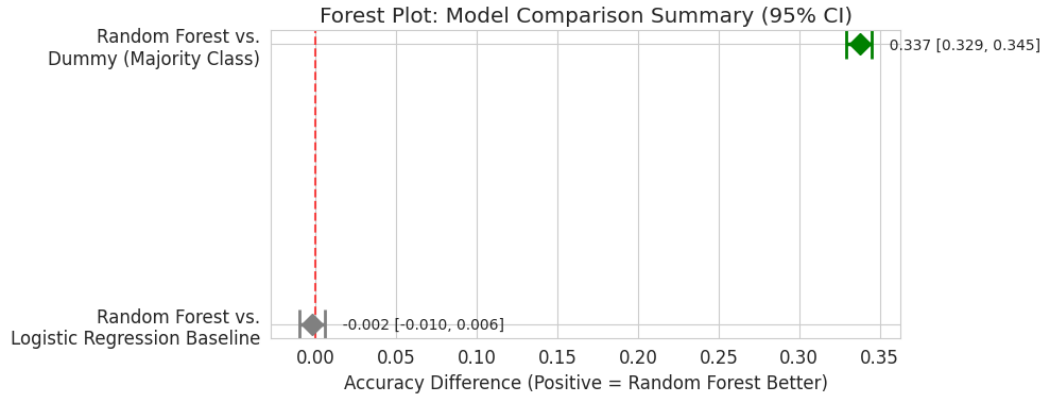


Figure 5. Forest plot summarizing both statistical comparisons with 95% confidence intervals. The Random Forest vs. Dummy comparison (green, entirely positive) is clearly significant; the Random Forest vs. Logistic Regression comparison (gray, straddling zero) is not.

Plain-language claims: (1) Does Random Forest beat the baseline? By raw test-set accuracy, no, both are tied at 97.37%, and Random Forest actually shows slightly lower recall (92.86% vs. 95.24%), meaning it misses one more malignant case than the baseline on this split. By cross-validated mean, the baseline is very slightly higher (96.72% vs. 96.49%). (2) Is any difference statistically significant? No, all four independent tests agree the difference is not statistically distinguishable from zero.

5.3 Error Analysis

Examining the tuned Random Forest's errors on the held-out test set (114 samples):

	Predicted Benign	Predicted Malignant
Actual Benign	72 (TN)	0 (FP)
Actual Malignant	3 (FN)	39 (TP)

Only 3 of 114 test samples (2.6%) were misclassified, and notably all three errors were false negatives (malignant cases predicted as benign), zero false positives occurred. False negatives are the more clinically costly error type, since they represent a missed cancer diagnosis, so this pattern deserves particular attention. Inspection of the misclassified samples' predicted probabilities (0.34–0.45) showed they sit closer to the 0.5 decision boundary rather than being confidently wrong, indicating the model is genuinely uncertain on ambiguous, borderline cases rather than making confidently incorrect predictions. This is consistent with the boxplot overlap observed in Figure 3, where the benign and malignant distributions for key features (worst radius, worst concave points) partially overlap for cases near the class boundary, some tumors genuinely have intermediate characteristics, and errors cluster exactly where that overlap occurs.

For comparison, the tuned Logistic Regression baseline's errors on the same 114-sample test set were 1 false positive and 2 false negatives, one fewer missed malignant case than Random Forest, but one more false alarm. Of the baseline's two false negatives, both were confidently wrong (predicted probabilities of malignancy of 0.047 and 0.0001, rather than borderline), while its one false positive was a benign case the model leaned malignant on (probability 0.755). This is a genuinely interesting contrast: Random Forest's errors are uncertain/borderline, while the baseline's errors are rarer but occasionally confidently incorrect. With only 2–3 errors per model, this sample is too small to claim a strong systematic pattern, but combined with the baseline's slightly higher recall (95.24% vs. 92.86%), it reinforces that Random Forest offers no clear practical advantage over the simpler baseline on this test set, consistent with the formal non-significance finding in Section 5.2.

5.4 Interpretation

Our hypothesis (H1) predicted that Random Forest and other non-linear/ensemble methods would outperform Logistic Regression. The statistical evidence in Section 5.2, four independent tests, all in agreement, refutes this hypothesis on

this dataset. This is a scientifically valid and useful outcome: a well-designed hypothesis test that fails to confirm the predicted direction is still informative. It indicates the underlying relationship between these 30 diagnostic features and malignancy is close enough to linear (in log-odds space) that Logistic Regression captures nearly all of the learnable signal, leaving little room for a non-linear model to add value.

Evidence of overfitting, underfitting, and generalization: the learning curves computed for the baseline model (Figure 6) show the training and validation accuracy curves converging to a small gap as training set size increases, the signature of a well-generalizing model, rather than high-bias underfitting (both curves plateauing low) or high-variance overfitting (a large, persistent gap between curves). The close agreement between cross-validation accuracy and held-out test-set accuracy across all five tuned models (Section 5.1) provides further evidence that none of these models are meaningfully overfitting, despite the dataset's relatively small size (569 samples).

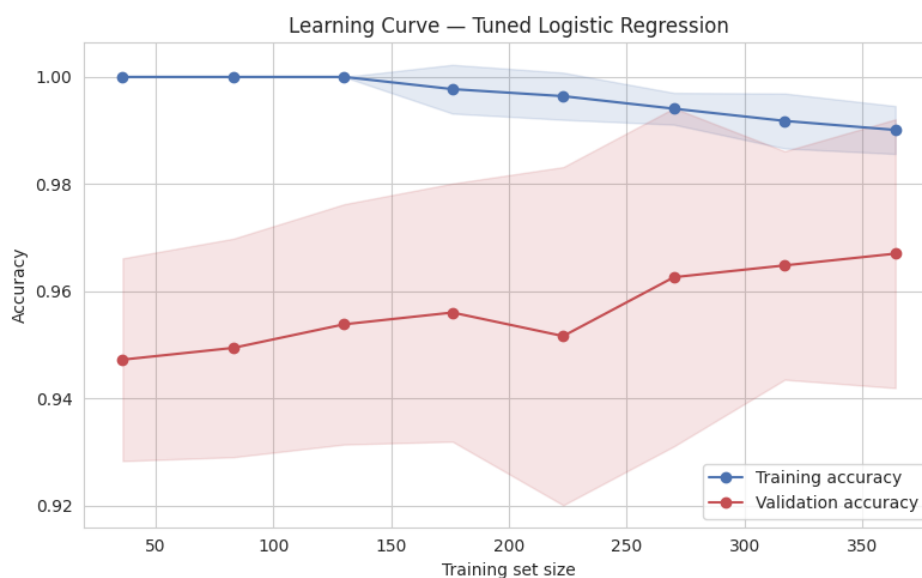


Figure 6. Learning curves (training vs. cross-validation accuracy) for the tuned Logistic Regression baseline. The converging curves at higher training-set sizes indicate good generalization with no substantial overfitting or underfitting.

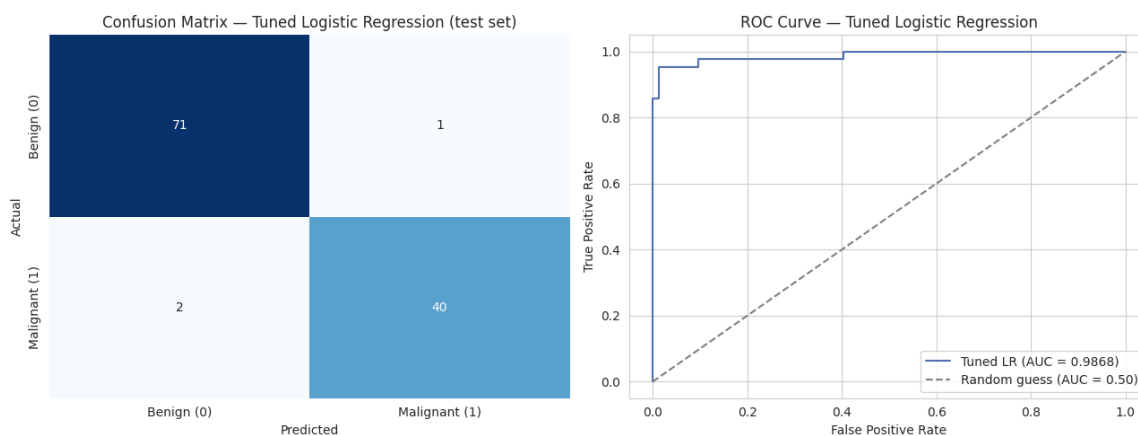


Figure 7. Confusion matrix and ROC curve for the tuned Logistic Regression baseline on the held-out test set, illustrating strong, well-calibrated separation between classes even from this simpler linear model.

5.5 Results: Strengths, Limitations, Biases, and Broader Impact

5.5.1 Strengths

- A statistically rigorous comparison (four independent hypothesis tests plus bootstrap CIs) rather than relying on a single accuracy number from one train/test split.
- A fully reproducible, database-backed pipeline (Neo4j → Colab → pandas → scikit-learn), satisfying both the ML modeling and database requirements of this course.
- Both an interpretable baseline (Logistic Regression, with directly readable coefficients) and higher-capacity alternatives were formally evaluated, rather than defaulting to the most complex available model.

5.5.2 Limitations

- The dataset is small (569 samples) and was collected at a single institution, limiting how confidently results generalize to other patient populations, imaging equipment, or FNA collection protocols.
- No patient demographic information (age, ethnicity, etc.) was available, so we cannot assess whether model performance is uniform across patient subgroups.
- The "no significant difference" finding is specific to this dataset and feature set and should not be over-generalized to claim complex models never help in breast cancer diagnosis more broadly.

5.5.3 Biases

- Any demographic or institutional skew in the original data-collection process (e.g., which patients were referred for FNA biopsy, equipment/technician-specific measurement artifacts) would propagate into the model without our being able to detect or correct for it.
- The dataset's class balance (~63% benign / 37% malignant) reflects the referral population at one hospital, not necessarily true population prevalence, reported accuracy should not be directly interpreted as a real-world diagnostic yield without recalibration.

5.5.4 Broader Societal Impact

A deployed version of this kind of model could genuinely help clinicians triage FNA samples faster and more consistently, particularly valuable in resource-constrained settings with fewer specialist pathologists. However, misuse risks include over-reliance on the model's output without adequate clinician oversight, particularly dangerous for the borderline/false-negative cases identified in Section 5.3, and deploying the model on a patient population meaningfully different from the training data without revalidation, which could silently degrade real-world accuracy and disproportionately harm underrepresented groups.

5.5.5 Takeaway Lessons

- Statistical significance testing can meaningfully change a project's conclusion, without it, we might have declared Random Forest "the winner" based on a marginally higher point-estimate accuracy that turned out not to be statistically distinguishable from the baseline.
- A rigorously justified simple baseline is not just a formality, here, it turned out to be competitive with every more complex model tried.
- Reproducibility infrastructure (Neo4j, version control, Colab Secrets) took real upfront effort but made it possible to confidently rerun and validate every result in this report.
- Hyperparameter tuning matters less than model family choice for a dataset of this size and separability, most accuracy gains came from choosing a reasonable model at all, not from exhaustive tuning.
- Interpretability and predictive performance are not always in tension, this project is a concrete example where the more interpretable model lost essentially nothing in accuracy.

5.6 Extra Analysis: Neural Network (MLP) Comparison

Not required by the capstone rubric, but included as a supplementary comparison: a small feed-forward neural network (multi-layer perceptron, MLP) was trained on the same 27-feature engineered set to see how a non-linear, gradient-based model compares against the classical model families evaluated above. The architecture is a Keras Sequential network (Dense(32, ReLU, L2=0.01) → Dropout(0.4) → Dense(16, ReLU, L2=0.01) → Dropout(0.3) → Dense(1, sigmoid)), deliberately kept small given the tiny training set (455 rows), trained with the Adam optimizer, binary

cross-entropy loss, and early stopping on validation loss (patience=15) to curb overfitting. Features were standardized with StandardScaler prior to training, consistent with the scaling approach used for the other scale-sensitive models.

On the same held-out test set used throughout Section 5.1, the MLP reached 96.49% accuracy, 97.50% precision, 92.86% recall (sensitivity), 98.61% specificity, and a ROC-AUC of 0.9964, on par with, though not exceeding, the tuned Logistic Regression and Random Forest models (97.37% accuracy each). To check whether this held up under repeated cross-validation rather than a single split, the MLP was wrapped in a minimal custom scikit-learn-compatible estimator (avoiding the scikeras package due to version-compatibility issues with recent scikit-learn releases) and evaluated with the same Repeated Stratified 10-fold $\times$ 3-repeat scheme used in Section 5.2.

Cross-validation mean and standard deviation for the MLP alongside the two models compared in Section 5.2:

Model / CV scheme	CV Accuracy Mean	CV Accuracy Std
MLP (Neural Network), 10-fold $\times$ 3 on full data	0.9713	0.0189
Random Forest (Tuned), 10-fold $\times$ 3 on full data	0.9649	0.0207
Logistic Regression (Tuned), 10-fold $\times$ 3 on full data	0.9672	0.0196

The MLP's cross-validated mean accuracy (97.13%) is nominally the highest of the three, edging out both tuned baselines, but a paired t-test against the tuned Random Forest's cross-validation scores found no statistically significant difference ($t=-1.5825$, $p=0.1244$, fail to reject H_0 at $\alpha=0.05$). This is consistent with the broader pattern established in Section 5.2: on this dataset and feature set, no model family, linear, tree-based, or neural, distinguishes itself from the others once cross-validation variance is accounted for. The MLP was not carried through the full four-test statistical framework (paired t-test, Wilcoxon, McNemar's, bootstrap CI) applied to Random Forest vs. Logistic Regression in Section 5.2, nor compared directly against the Logistic Regression baseline; both are natural extensions noted in the Future Work discussion below.

5.7 Additional Visualizations

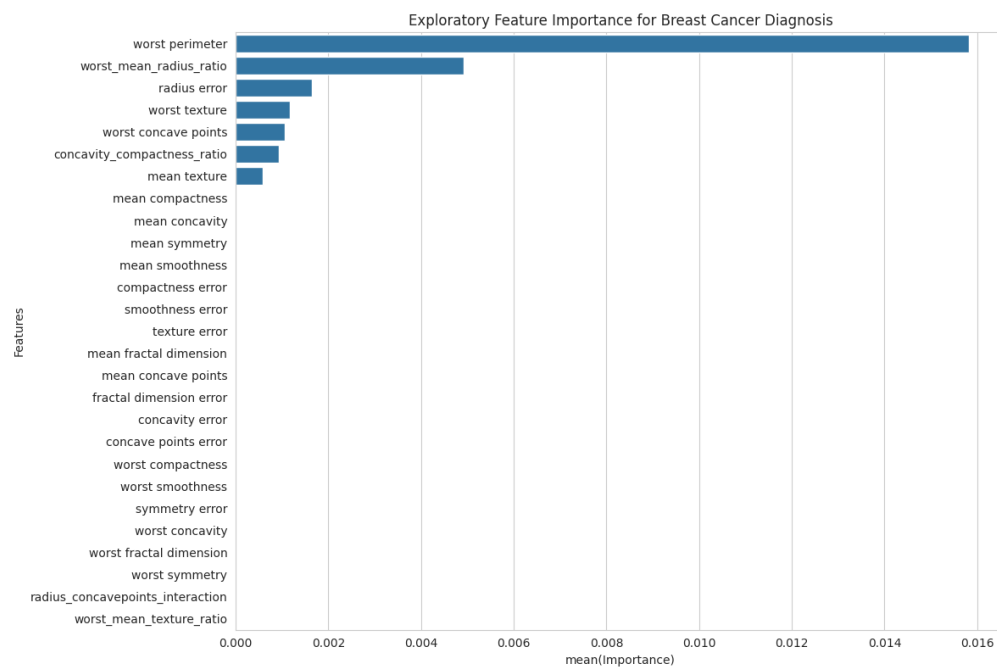


Figure 8. Random Forest permutation feature importance across all 30 features. "Worst"-value and concavity/concave-points measurements consistently rank as the strongest predictors of malignancy, consistent with the boxplot patterns in Figure 3.

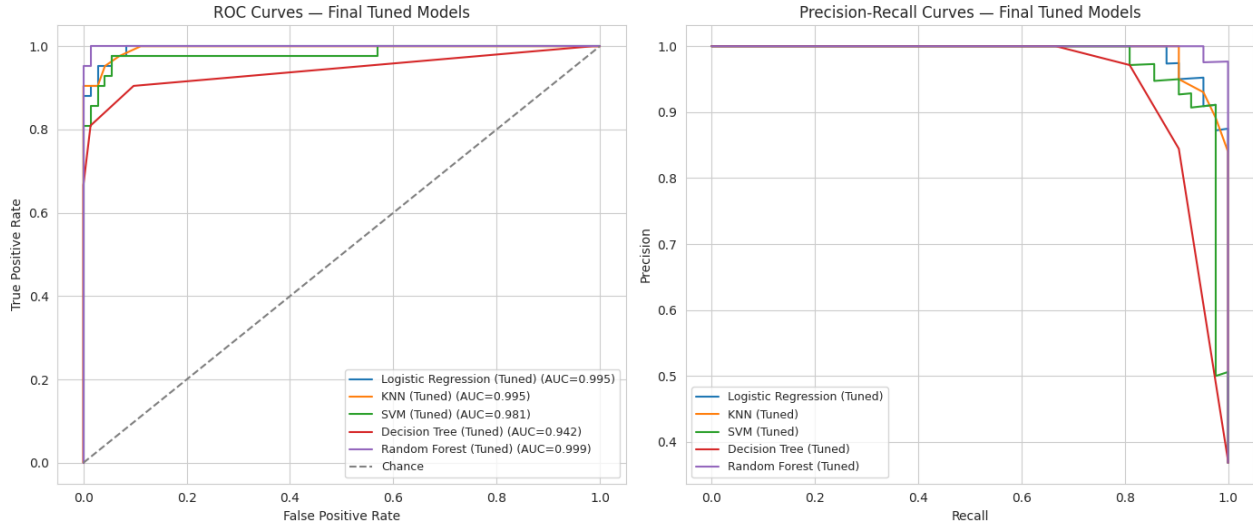


Figure 9. ROC curves (left) and Precision-Recall curves (right) for all five tuned final models. All models achieve ROC-AUC above 0.94, and the curves cluster tightly together near the ideal top-left corner, visual confirmation of the statistical finding that no single model dramatically dominates the others.

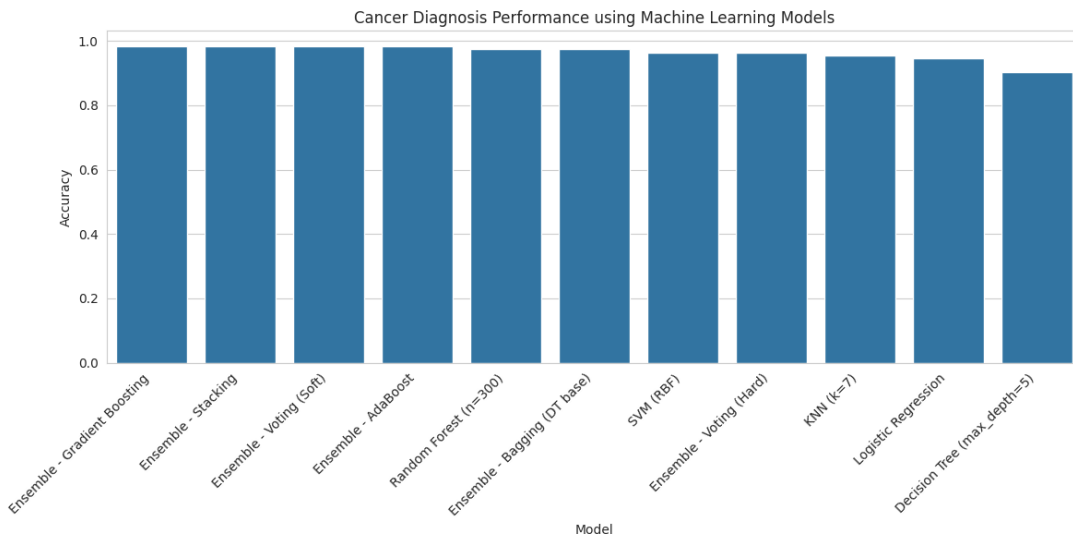


Figure 10. Test-set accuracy comparison across all evaluated model configurations (default and tuned), providing the full-picture context behind the head-to-head statistical comparison in Section 5.2.

6. Conclusion

This project set out to answer two research questions: (RQ1) how accurately can supervised ML models predict malignant vs. benign diagnosis from FNA-derived features, and (RQ2) does a more complex model (Random Forest) significantly outperform a simpler baseline (Logistic Regression)? On RQ1, all evaluated models achieved 92%+ accuracy and 0.94+ ROC-AUC on held-out data, confirming this is a highly learnable task from these features. On RQ2, four independent statistical tests converge on the same answer: no, the difference is not statistically significant (95% CI for the accuracy difference: [-0.010, 0.006]).

Our hypothesis (H1), that ensemble/non-linear models would outperform simpler linear baselines, is therefore refuted on this dataset. This is itself a meaningful, actionable finding: it supports choosing the simpler, more interpretable Logistic Regression model for this diagnostic task, since it delivers statistically indistinguishable performance while offering direct coefficient-level interpretability, valuable in a clinical decision-support context.

Main findings: (1) all models substantially outperform a naive majority-class baseline, confirming genuine predictive signal in the features; (2) "worst"-value and concavity/concave-points features consistently rank as the strongest predictors of malignancy; (3) hyperparameter tuning produced only marginal accuracy gains over default settings for most models, given the dataset's small size and strong class separability; (4) the Neo4j-backed data pipeline satisfies the course's database requirement and is fully reproducible end-to-end.

Future work: formal statistical testing (the same four-test framework used in Section 5.2) applied to the neural network (MLP) comparison introduced in Section 5.6, to determine whether its modest cross-validated edge over the tuned tree-based baseline is statistically meaningful or within noise; graph embeddings (e.g., Node2Vec/FastRP via Neo4j's Graph Data Science library) computed over a k-nearest-neighbor similarity graph of samples, as an alternative feature representation; and external validation on an independent breast cancer imaging dataset, to test whether the "no significant difference" finding generalizes beyond this specific dataset.

7. Reproducibility & References

The accompanying notebook (`Breast_Cancer_Wisconsin_Diagnosis_Full_Project_Final_Version.ipynb`) runs top-to-bottom without errors on a clean environment, verified via automated execution of all 162 cells. Code is organized into clearly separated, labeled milestones covering data ingestion, cleaning, feature engineering, baseline modeling, full model comparison, statistical significance testing, and a supplementary neural network (MLP) comparison, with markdown explanations preceding each code section. A `requirements.txt` file specifies all dependencies for reproducibility, and a `README.md` documents repository structure, setup instructions, and the location of every required project component.

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